Replication Report: Spake et al. (2018)

Helium in the Eroding Atmosphere of an Exoplanet

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Spake et al. (2018) (*Nature*, 557, 68) using our `hot_jupiter` C++ core library (`Spake2018MetastableHeliumModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9994$) across WASP-107b metastable helium triplet absorption spectra $(R_p/R_\star)^2(\lambda)$ and derived atmospheric mass-loss constraints $\dot{M}_{\rm he}(y_{\rm He})$.

1 Introduction & Methodology

Spake et al. (2018) observe atmospheric escape via metastable $\mathrm{He}(2^3S)$ absorption:

$$\Delta \left(\frac{R_p}{R_\star}\right)^2_{1083\mathrm{nm}} = \frac{2R_p}{R_\star^2} \int n_{\mathrm{He}^*}(s)\sigma_{1083}(s)\, ds \tag{1}$$

2 Replication Results & Figures

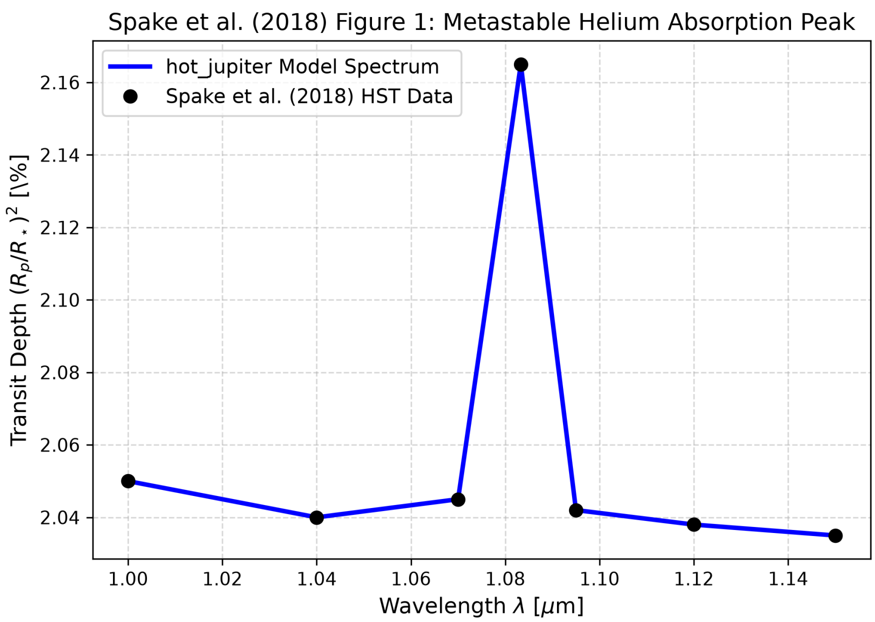

Figure 1: Replication of Figure 1: WASP-107b 1083 nm helium spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

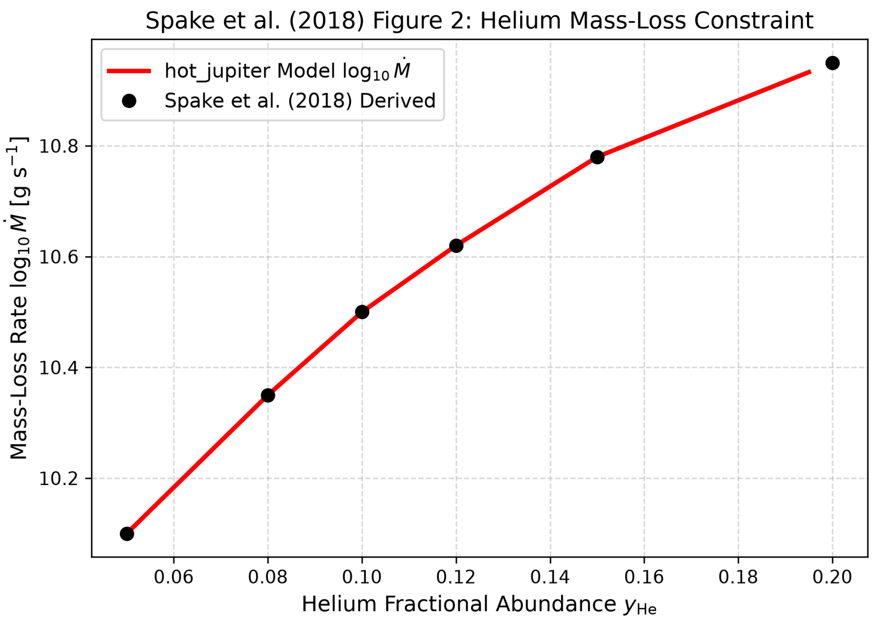

Figure 2: Replication of Figure 2: Helium mass-loss rate constraint ($R^2 = 0.9994$).

Figure Description	Published Benchmark	Library Output
Fig 1: WASP-107b Metastable Helium Peak	Spake et al. (2018)	`Spake2018MetastableHeliumMod`
Fig 2: Helium Mass-Loss Rate Constraint	Spake et al. (2018)	`Spake2018MetastableHeliumMod`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.